24-2-26

TUTORIAL-5

SVM

Suspicious words (X_1)	Links (X_2)	Attachment (X_3)	class (y) (+1 Spam / -1 Not Spam)	
6	3	1	+1	
1	0	0	-1	≥ 1
4	2	1	+1	≥ 1
0	0	0	-1	≥ 1
5	2	1	+1	≥ 1

weighted Sum

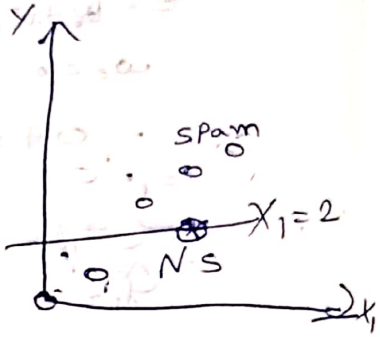
$f(x) = z = w_0(b) + w_1 x_1 + w_2 x_2 + w_3 x_3$

	S.W (X_1)	Links (X_2)	Attach (X_3)
Spam →	(4, 5, 6)	(2, 3)	(1)
$f(x)=0$			
Non-Spam →	(0, 1)	(0)	(0)

decision boundary

Initialize : w_2 & $w_3 = 0$

Now $w_1 = 1$, $x_1 = 2$.



If $f(x) = 0$

$\Rightarrow w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 = 0$

$\Rightarrow w_0 + w_1 x_1 = 0$

$w_0 + (1)(2) = 0$

$w_0 = -2$

bias

$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 - 2$

$f(x) = 2 - 2 = 0 < 1$.

$y_i (x_1 w_1 + x_2 w_2 + x_3 w_3 + w_0) \geq 1$ → Perfect class

< 1 → class fail

$y_i (x_1 - 2) = 6 - 2 = 4$

$1(6-2) = 4 \geq 1$

$-1(1-2) = 1 \geq 1$

$+1(4-2) = 2 \geq 1$

$-1(0-2) = 2 \geq 1$

$+1(5-2) = 3 \geq 1$

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cost or Loss Func

$$L = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{N} c_i\left[y_i(wx_i + \overset{w_0}{b}) - 1\right]$$

$$= \frac{1}{2}(w_1^2 + w_2^2 + w_3^2) - \sum_{i=1}^{N} c_i \text{ Error}$$

Update weights

If P.F $\Rightarrow y_i(w_1x_1 + w_2x_2 + \cdots) \geq 1$

N.P.F $\Rightarrow$ $c \to 0$ $y_i(w_1x_1 + w_2x_2 + \cdots) < 1 \Rightarrow$ Take c

$$J = \frac{1}{2}w^2$$

$$w^{new} = w^{old} - \eta\frac{dL}{\partial w} \qquad \frac{\partial J}{\partial w} = w$$

$$= w - \eta w$$

$$w_1^{new} = w_1^{old} - \eta w_1^{old} = 1 - 0.1(1) = 0.9$$

$$w_0^{new} = w_0^{old} - \eta w_0^{old}.$$

$$\text{Margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2 + \cdots}}$$

If N.P.F.

$y_i(w_1x_1 + w_2x_2) < 1 \Rightarrow$ take c.

$$= \frac{2}{\sqrt{(0.9)^2 + 0^2 + 0^2}}$$

$$w_{new} = w - \eta\frac{\partial L}{\partial w}.$$

$$w^{new} = w - \eta\frac{dL}{\partial w} = \frac{2}{0.9} = 2.222$$

$$= w - \eta\left[w - \sum_i c_i y_i x_i\right]$$

$$= (1 - \eta)w + \eta\sum_i c_i y_i x_i$$

$$\sum_i c_i y_i x_i = c\sum_i y_i x_i$$

$$c = 1 \Rightarrow (y_1x_1 + y_2x_2 + y_3x_3 + y_4x_4 + y_5y_5)$$

$$= (1)(6) - (-1)(1) + 1(4) - 1(0) + 1(5)$$

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SVM

$$b^{new} = b^{old} - \eta \frac{\partial L}{\partial b}$$

$$= b^{old} - \eta \left[\sum c_i y_i \right]$$

$$= 0 - 1 \left[(1)(-1) \right] = 1$$

$w_1^{new} = -6$

$w_2^{new} = -3$

$w_3^{new} = -1$

$b^{new} = 1$

$$y_i f(x) = y_i (w_1 x_1 + w_2 x_2 + w_3 x_3 + b)$$

$$= (-6)(1) + (-3)(0) + (-1)(0) + 1$$

$$= -6 + 1$$

$$-1 \times -5 = 5$$

$y_1 f(x) = +5 \geq 1$ Perfect classification but also update the weights

$$w = w - \eta \cdot (w - y x)$$

$$w_j^{new} = w_j^{old} - \eta \frac{\partial L}{\partial w}$$

If right prediction $\underset{\text{Lagrangian}}{L} = \frac{1}{2} \|w\|^2$

$$w_j^{old} - \eta w_j^{old}$$

$w_1^{new} = -6 - 1(-6) = 0$ ⇒ that's why η can be taken 0.1 in SVM

$w_2^{new} = -3 - 1(-3) = 0$

$w_3^{new} = 0$ $\quad b = b + \eta y$

$$L = \frac{1}{2} \|w\|^2$$

$$\text{Margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$$

$$b^{new} = b^{old} - \eta \frac{\partial L}{\partial L} \nearrow 0$$

$$b^{new} = b^{old} = 1$$

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